

Design Verification of Steel Connections Using Osdag

Ashu Yadav
B.Tech Civil Engineering
IIT Roorkee

Abstract

This report presents the manual verification of steel connections designed using Osdag. All calculations are performed as per IS 800:2007. The objective is to validate software outputs and identify assumptions, inconsistencies, and possible improvements in the design philosophy.

Contents

1	Beam-to-Beam Cover Plate Bolted Connection	5
1.1	Introduction	5
1.2	Input Data	5
1.3	Member Capacity	5
1.3.1	Axial Capacity	5
1.3.2	Shear Capacity	5
1.3.3	Moment Capacity	6
1.4	Load Adjustment	6
1.5	Force Distribution	6
1.6	Bolt Design	6
1.6.1	Assumptions	6
1.6.2	Bolt Shear Capacity	6
1.6.3	Bolt Bearing Capacity	6
1.6.4	Governing Capacity	6
1.7	Plate Design Check	7
1.8	Bolt Spacing Check	7
1.9	Comparison with Osdag	7
1.10	Observations	7
1.11	Bugs and Recommendations	7
1.12	Conclusion	7
2	Beam-to-Column End Plate Connection	8
2.1	Introduction	8
2.2	Input Data	8
2.3	Load Adjustment (IS 800 Cl. 10.7)	8
2.4	Bolt Design	9
2.4.1	Assumptions	9
2.4.2	Bolt Shear Capacity	9
2.4.3	Bolt Bearing Capacity	9
2.4.4	Governing Capacity	9
2.5	Bolt Force Check	9
2.5.1	Shear per Bolt	9
2.5.2	Tension in Bolt	9
2.5.3	Interaction Check	9
2.6	Prying Force	10
2.7	End Plate Design	10
2.7.1	Moment at Critical Section	10
2.7.2	Required Thickness	10

2.8	Weld Design	10
2.8.1	Required Weld Size	10
2.9	Comparison with Osdag	10
2.10	Observations	10
2.11	Bugs and Recommendations	11
2.12	Conclusion	11
3	Column-to-Column Cover Plate Bolted Connection	12
3.1	Introduction	12
3.2	Input Data	12
3.3	Load Adjustment (IS 800 Cl. 10.7)	12
3.4	Force Distribution	12
3.5	Flange Bolt Design	13
3.5.1	Bolt Properties	13
3.5.2	Shear Capacity	13
3.5.3	Bearing Capacity	13
3.5.4	Governing Capacity	13
3.5.5	Long Joint Reduction	13
3.6	Flange Bolt Force Check	13
3.7	Web Bolt Design	13
3.7.1	Bolt Capacity	13
3.7.2	Resultant Force	13
3.8	Plate Design	14
3.8.1	Flange Plate	14
3.8.2	Web Plate	14
3.9	Member Capacity Check	14
3.9.1	Flange	14
3.9.2	Web	14
3.10	Comparison with Osdag	14
3.11	Observations	14
3.12	Bugs and Recommendations	14
3.13	Conclusion	15
4	Column-to-Column Cover Plate Welded Connection	16
4.1	Introduction	16
4.2	Input Data	16
4.3	Load Adjustment (IS 800 Cl. 10.7)	16
4.4	Force Distribution	16
4.5	Flange Weld Design	17
4.5.1	Weld Size Limits	17
4.5.2	Throat Thickness	17
4.5.3	Effective Length	17
4.5.4	Weld Stress	17
4.5.5	Weld Strength	17
4.6	Web Weld Design	17
4.6.1	Weld Size	17
4.6.2	Effective Length	17
4.6.3	Resultant Weld Stress	17

4.6.4	Weld Strength	18
4.7	Plate Design	18
4.7.1	Flange Plate	18
4.7.2	Web Plate	18
4.8	Member Capacity Check	18
4.8.1	Flange	18
4.8.2	Web	18
4.9	Comparison with Osdag	18
4.10	Observations	18
4.11	Bugs and Recommendations	19
4.12	Conclusion	19
5	Column-to-Column End Plate Bolted Connection	20
5.1	Introduction	20
5.2	Input Data	20
5.3	Load Adjustment (IS 800 Cl. 10.7)	20
5.4	Bolt Design	20
5.4.1	Bolt Details	20
5.4.2	Bolt Shear Capacity	21
5.4.3	Bolt Bearing Capacity	21
5.4.4	Governing Capacity	21
5.4.5	Shear per Bolt	21
5.5	Bolt Tension and Prying Force	21
5.5.1	Tension due to Moment	21
5.5.2	Prying Force	21
5.5.3	Total Tension	21
5.5.4	Tension Capacity	21
5.6	Bolt Spacing Check	21
5.7	End Plate Design	22
5.7.1	Plate Thickness	22
5.7.2	Moment Capacity	22
5.8	Member Capacity Check	22
5.9	Comparison with Osdag	22
5.10	Observations	22
5.11	Bugs and Recommendations	22
5.12	Conclusion	22
6	Overall Observations	23
7	Recommendations to Design Philosophy and Bugs	24
7.1	Lack of Transparency in Load Modification	24
7.2	Hidden Intermediate Calculations	24
7.3	Design Philosophy Concerns	24
7.4	Software Improvement Suggestions	24
8	Conclusion	25

Chapter 1

Beam-to-Beam Cover Plate Bolted Connection

1.1 Introduction

This report presents the manual verification of a beam-to-beam cover plate bolted connection designed using Osdag. All calculations are performed as per IS 800:2007 to validate the software-generated results.

1.2 Input Data

- Beam Section: MB 250
- Material: Fe 410 ($f_y = 250$ MPa, $f_u = 410$ MPa)
- Factored Moment: 25 kNm
- Factored Shear: 15 kN
- Bolt Diameter: 10 mm
- Bolt Grade: 6.8 ($f_{ub} = 400$ MPa)

1.3 Member Capacity

1.3.1 Axial Capacity

$$T_d = \frac{A_g f_y}{\gamma_{m0}} = \frac{4750 \times 250}{1.1 \times 10^3} = 1079.55 \text{ kN} \quad (1.1)$$

1.3.2 Shear Capacity

$$V_d = \frac{A_v f_y}{\sqrt{3} \gamma_{m0}} = 203.71 \text{ kN} \quad (1.2)$$

1.3.3 Moment Capacity

$$M_d = \frac{Z_p f_y}{\gamma_{m0}} = 105.68 \text{ kNm} \quad (1.3)$$

1.4 Load Adjustment

As per IS 800 Clause 10.7:

- Adjusted Shear = 30.56 kN
- Adjusted Moment = 52.84 kNm

1.5 Force Distribution

Force is distributed between flange and web:

- Web Force (A_w) = 105.85 kN
- Flange Force (F_f) = 287.24 kN

1.6 Bolt Design

1.6.1 Assumptions

- Bolts are in single shear
- Threads are in shear plane
- Partial safety factor $\gamma_{mb} = 1.25$

1.6.2 Bolt Shear Capacity

Net area of bolt, $A_{nb} = 58 \text{ mm}^2$

$$V_{dsb} = \frac{f_{ub} \cdot A_{nb}}{\sqrt{3} \cdot \gamma_{mb}} = \frac{400 \times 58}{\sqrt{3} \times 1.25 \times 1000} = 16.07 \text{ kN} \quad (1.4)$$

1.6.3 Bolt Bearing Capacity

$$V_{dpb} = \frac{2.5 \cdot k_b \cdot d \cdot t \cdot f_u}{\gamma_{mb}} = 59.45 \text{ kN} \quad (1.5)$$

1.6.4 Governing Capacity

$$V_{db} = \min(16.07, 59.45) = 16.07 \text{ kN} \quad (1.6)$$

1.7 Plate Design Check

Plate thickness is checked against yielding:

$$T = \frac{F}{b \cdot f_y} \quad (1.7)$$

Plate is found to be safe under applied forces.

1.8 Bolt Spacing Check

- Minimum pitch = $2.5d = 25$ mm
- Minimum edge distance = $1.5d = 15$ mm

Provided spacing satisfies IS 800 requirements.

1.9 Comparison with Osdag

Parameter	Osdag	Manual	Difference (%)
Bolt Capacity (kN)	16.07	16.07	0
Flange Force (kN)	287.24	287.24	0

1.10 Observations

- Manual results match Osdag outputs with negligible deviation.
- Shear governs bolt design.
- IS 800 provisions provide conservative estimates.

1.11 Bugs and Recommendations

- Lack of transparency in k_b calculation in Osdag
- Rounding differences observed in intermediate steps
- Recommendation: Include detailed intermediate steps in Osdag report

1.12 Conclusion

Manual verification confirms that the design generated by Osdag is accurate and follows IS 800:2007 provisions. The connection is safe under the given loading conditions.

Chapter 2

Beam-to-Column End Plate Connection

2.1 Introduction

This report presents the manual verification of a beam-to-column end plate connection designed using Osdag. The verification is carried out as per IS 800:2007 provisions.

2.2 Input Data

- Beam Section: MB 250
- Column Section: HB 450
- Material: Fe 410 ($f_y = 250$ MPa, $f_u = 410$ MPa)
- Applied Moment: 5 kNm
- Applied Shear: 5 kN
- Bolt: 8 mm diameter, Grade 4.6 ($f_{ub} = 400$ MPa)

2.3 Load Adjustment (IS 800 Cl. 10.7)

Minimum load requirements:

$$M_u = 0.5M_d = 0.5 \times 105.68 = 52.84 \text{ kNm} \quad (2.1)$$

$$V_u = 0.15V_d = 18.33 \text{ kN} \quad (2.2)$$

Adopted Loads:

- Moment = 52.84 kNm
- Shear = 18.33 kN

2.4 Bolt Design

2.4.1 Assumptions

- Bolts in single shear
- Threads in shear plane
- $\gamma_{mb} = 1.25$

2.4.2 Bolt Shear Capacity

Net area $A_{nb} = 36.6 \text{ mm}^2$

$$V_{dsb} = \frac{f_{ub} \cdot A_{nb}}{\sqrt{3} \cdot \gamma_{mb}} = \frac{400 \times 36.6}{\sqrt{3} \times 1.25 \times 1000} = 6.76 \text{ kN} \quad (2.3)$$

2.4.3 Bolt Bearing Capacity

$$k_b = \min \left(\frac{e}{3d_0}, \frac{p}{3d_0} - 0.25, \frac{f_u}{f_{ub}}, 1.0 \right) \quad (2.4)$$

$$k_b = \min \left(\frac{15}{24}, \frac{30}{24} - 0.25, \frac{410}{400}, 1.0 \right) = 0.62 \quad (2.5)$$

$$V_{dpb} = \frac{2.5 \cdot k_b \cdot d \cdot t \cdot f_u}{\gamma_{mb}} = \frac{2.5 \times 0.62 \times 8 \times 14 \times 410}{1.25 \times 1000} = 56.94 \text{ kN} \quad (2.6)$$

2.4.4 Governing Capacity

$$V_{db} = \min(6.76, 56.94) = 6.76 \text{ kN} \quad (2.7)$$

2.5 Bolt Force Check

2.5.1 Shear per Bolt

$$V_{sb} = \frac{V_u}{n} = \frac{18.33}{24} = 0.76 \text{ kN} \quad (2.8)$$

2.5.2 Tension in Bolt

$$T_b = 9.65 \text{ kN} \quad (2.9)$$

2.5.3 Interaction Check

$$\left(\frac{V_{sb}}{V_{db}} \right)^2 + \left(\frac{T_b}{T_{db}} \right)^2 = \left(\frac{0.76}{6.76} \right)^2 + \left(\frac{9.65}{10.54} \right)^2 = 0.85 < 1 \quad (2.10)$$

Safe

2.6 Prying Force

$$Q = 0 \quad (2.11)$$

Plate thickness is sufficient to prevent prying action.

2.7 End Plate Design

2.7.1 Moment at Critical Section

$$M_{cr} = T_b \cdot l_v = 9.65 \times 8.5 \times 10^{-3} = 0.08 \text{ kNm} \quad (2.12)$$

2.7.2 Required Thickness

$$t_p = \sqrt{\frac{4M_{cr}}{b_e(f_y/\gamma_{m0})}} = 6.8 \text{ mm} \quad (2.13)$$

Provided thickness = 14 mm (Safe)

2.8 Weld Design

2.8.1 Required Weld Size

$$t_w = \frac{V_u}{f_u \cdot L_w} \times \sqrt{3}\gamma_{mw} = 0.39 \text{ mm} \quad (2.14)$$

Provided weld = 6 mm (Safe)

2.9 Comparison with Osdag

Parameter	Osdag	Manual	Difference (%)
Bolt Capacity (kN)	6.76	6.76	0
Shear per Bolt (kN)	0.76	0.76	0
Tension (kN)	9.65	9.65	0

2.10 Observations

- Manual calculations match Osdag results accurately.
- Shear governs bolt design.
- Load adjustment significantly increases design forces.

2.11 Bugs and Recommendations

- Osdag automatically modifies loads as per IS 800 Clause 10.7 without clear visibility.
- Plate thickness optimization is not transparent.
- Recommendation: Provide step-by-step calculation details in output.

2.12 Conclusion

Manual verification confirms that the Osdag design is accurate and complies with IS 800:2007. The connection is safe under the given loading conditions.

Chapter 3

Column-to-Column Cover Plate Bolted Connection

3.1 Introduction

This report presents the manual verification of a column-to-column cover plate bolted connection designed using Osdag. The verification is carried out as per IS 800:2007 provisions.

3.2 Input Data

- Column Section: HB 400
- Material: Fe 410 ($f_y = 250$ MPa, $f_u = 410$ MPa)
- Applied Moment = 20 kNm
- Applied Shear = 10 kN
- Applied Axial Load = 5 kN

3.3 Load Adjustment (IS 800 Cl. 10.7)

$$P_u = 672.27 \text{ kN} \quad (3.1)$$

$$V_u = 40 \text{ kN} \quad (3.2)$$

$$M_u = 177.27 \text{ kNm} \quad (3.3)$$

Note: Loads are significantly increased as per IS 800 minimum design requirements.

3.4 Force Distribution

- Flange Force = 580.52 kN
- Web Force = 232.42 kN

3.5 Flange Bolt Design

3.5.1 Bolt Properties

- Diameter = 22 mm
- Grade = 3.6 ($f_{ub} = 330$ MPa)
- Net Area = 303 mm²

3.5.2 Shear Capacity

$$V_{dsb} = \frac{f_{ub} \cdot A_{nb}}{\sqrt{3} \cdot \gamma_{mb}} = \frac{330 \times 303}{\sqrt{3} \times 1.25 \times 1000} = 46.18 \text{ kN} \quad (3.4)$$

3.5.3 Bearing Capacity

$$V_{dpb} = \frac{2.5 \cdot k_b \cdot d \cdot t \cdot f_u}{\gamma_{mb}} = 116.85 \text{ kN} \quad (3.5)$$

3.5.4 Governing Capacity

$$V_{db} = 46.18 \text{ kN} \quad (3.6)$$

3.5.5 Long Joint Reduction

$$V_{rd} = 0.9 \times 46.18 = 41.57 \text{ kN} \quad (3.7)$$

3.6 Flange Bolt Force Check

$$V_{res} = 41.47 \text{ kN} \quad (3.8)$$

Since $V_{res} < V_{rd}$, design is safe.

3.7 Web Bolt Design

3.7.1 Bolt Capacity

$$V_{db} = 83.72 \text{ kN} \quad (3.9)$$

3.7.2 Resultant Force

$$V_{res} = 75.87 \text{ kN} \quad (3.10)$$

Safe since $75.87 < 83.72$

3.8 Plate Design

3.8.1 Flange Plate

- Thickness = 14 mm
- Area = 3500 mm² (Safe)

3.8.2 Web Plate

- Thickness = 8 mm
- Area = 4080 mm² (Safe)

3.9 Member Capacity Check

3.9.1 Flange

$$T_d = 721.59 \text{ kN} > 580.52 \text{ kN (Safe)} \quad (3.11)$$

3.9.2 Web

$$T_d = 729.0 \text{ kN} > 232.42 \text{ kN (Safe)} \quad (3.12)$$

3.10 Comparison with Osdag

Parameter	Osdag	Manual	Difference (%)
Bolt Capacity (kN)	46.18	46.18	0
Flange Force (kN)	580.52	580.52	0
Web Force (kN)	232.42	232.42	0

3.11 Observations

- Significant increase in design loads due to IS 800 Clause 10.7.
- Flange carries majority of moment.
- Long joint reduction affects bolt capacity.

3.12 Bugs and Recommendations

- Load amplification is not clearly visible in initial inputs.
- Reduction factors are applied internally without explanation.
- Recommendation: Improve transparency in intermediate calculations.

3.13 Conclusion

Manual verification confirms that the Osdag design is accurate and complies with IS 800:2007 provisions. The connection is safe under combined loading.

Chapter 4

Column-to-Column Cover Plate Welded Connection

4.1 Introduction

This report presents the manual verification of a column-to-column cover plate welded connection designed using Osdag. The connection is subjected to combined axial force, shear force, and bending moment as per IS 800:2007.

4.2 Input Data

- Column Section: HB 450
- Material: Fe 410 ($f_y = 250$ MPa, $f_u = 410$ MPa)
- Applied Moment = 10 kNm
- Applied Shear = 10 kN
- Applied Axial Load = 0 kN

4.3 Load Adjustment (IS 800 Cl. 10.7)

From Osdag (Page 4–5):

$$P_u = 756.82 \text{ kN} \quad (4.1)$$

$$V_u = 40 \text{ kN} \quad (4.2)$$

$$M_u = 221.59 \text{ kNm} \quad (4.3)$$

Note: Loads are increased significantly as per minimum design requirements.

4.4 Force Distribution

From Page 5–6:

- Web Force = 282.37 kN
- Flange Force = 627.45 kN

4.5 Flange Weld Design

4.5.1 Weld Size Limits

- Minimum weld size = 5 mm
- Maximum weld size = 13.7 mm
- Provided weld = 12 mm

4.5.2 Throat Thickness

$$t_t = 0.7 \times 12 = 8.4 \text{ mm} \quad (4.4)$$

4.5.3 Effective Length

$$l_{eff} = 695 \text{ mm} \quad (4.5)$$

4.5.4 Weld Stress

$$\text{Stress} = \frac{F_f}{l_{eff}} = \frac{627.45 \times 10^3}{695} = 906.72 \text{ N/mm} \quad (4.6)$$

4.5.5 Weld Strength

$$f_w = \frac{t_t \cdot f_u}{\sqrt{3} \cdot \gamma_{mw}} = \frac{8.4 \times 410}{\sqrt{3} \times 1.25} = 1590.72 \text{ N/mm} \quad (4.7)$$

Safe since $906.72 < 1590.72$

4.6 Web Weld Design

4.6.1 Weld Size

- Provided weld = 8 mm
- Throat thickness = 5.6 mm

4.6.2 Effective Length

$$l_{eff} = 595 \text{ mm} \quad (4.8)$$

4.6.3 Resultant Weld Stress

From Page 10:

$$R_w = 522.3 \text{ N/mm} \quad (4.9)$$

4.6.4 Weld Strength

$$f_w = \frac{5.6 \times 410}{\sqrt{3} \times 1.25} = 1060.48 \text{ N/mm} \quad (4.10)$$

Safe since $522.3 < 1060.48$

4.7 Plate Design

4.7.1 Flange Plate

- Width = 215 mm
- Thickness = 18 mm
- Area = 3870 mm² (Safe)

4.7.2 Web Plate

- Height = 360 mm
- Thickness = 10 mm
- Area = 7200 mm² (Safe)

4.8 Member Capacity Check

4.8.1 Flange

$$T_d = 778.41 \text{ kN} > 627.45 \text{ kN (Safe)} \quad (4.11)$$

4.8.2 Web

$$T_d = 941.25 \text{ kN} > 282.37 \text{ kN (Safe)} \quad (4.12)$$

4.9 Comparison with Osdag

Parameter	Osdag	Manual	Difference (%)
Flange Force (kN)	627.45	627.45	0
Web Force (kN)	282.37	282.37	0
Flange Weld Stress	906.72	906.72	0

4.10 Observations

- Significant load amplification observed due to IS 800 Clause 10.7.
- Weld design governs the connection behavior.
- Flange carries major portion of moment.

4.11 Bugs and Recommendations

- Load modification is not explicitly shown in initial stage.
- Weld stress distribution assumptions are not clearly explained.
- Recommendation: Provide intermediate weld force calculations in output.

4.12 Conclusion

Manual verification confirms that the welded connection design is safe and complies with IS 800:2007 provisions.

Chapter 5

Column-to-Column End Plate Bolted Connection

5.1 Introduction

This report presents the manual verification of a column-to-column end plate bolted connection designed using Osdag. The connection is subjected to combined axial load, shear force, and bending moment as per IS 800:2007.

5.2 Input Data

- Column Section: HB 300
- Material: Fe 410 ($f_y = 250$ MPa, $f_u = 410$ MPa)
- Applied Moment = 165 kNm
- Applied Shear = 25 kN
- Applied Axial Load = 8 kN

5.3 Load Adjustment (IS 800 Cl. 10.7)

From Osdag (Page 4–5):

$$P_u = 223.68 \text{ kN} \quad (5.1)$$

$$V_u = 40 \text{ kN} \quad (5.2)$$

$$M_u = 165 \text{ kNm} \quad (5.3)$$

Note: Axial load increased significantly from 8 kN to 223.68 kN.

5.4 Bolt Design

5.4.1 Bolt Details

- Diameter = 27 mm

- Grade = 9.8 ($f_{ub} = 900$ MPa)
- Hole diameter = 30 mm
- Number of bolts = 6

5.4.2 Bolt Shear Capacity

$$V_{dsb} = \frac{f_{ub} \cdot A_{nb}}{\sqrt{3} \cdot \gamma_{mb}} = \frac{900 \times 459}{\sqrt{3} \times 1.25 \times 1000} = 190.8 \text{ kN} \quad (5.4)$$

5.4.3 Bolt Bearing Capacity

$$V_{dpb} = \frac{2.5 \cdot k_b \cdot d \cdot t \cdot f_u}{\gamma_{mb}} = 657.12 \text{ kN} \quad (5.5)$$

5.4.4 Governing Capacity

$$V_{db} = 190.8 \text{ kN} \quad (5.6)$$

5.4.5 Shear per Bolt

$$V_{sb} = \frac{V_u}{n} = \frac{40}{6} = 6.67 \text{ kN} \quad (5.7)$$

Safe since $6.67 < 190.8$

5.5 Bolt Tension and Prying Force

5.5.1 Tension due to Moment

$$T_1 = 282.55 \text{ kN} \quad (5.8)$$

5.5.2 Prying Force

$$Q = 23.94 \text{ kN} \quad (5.9)$$

5.5.3 Total Tension

$$T_b = T_1 + Q = 306.49 \text{ kN} \quad (5.10)$$

5.5.4 Tension Capacity

$$T_{db} = 297.43 \text{ kN} \quad (5.11)$$

Safe (as per Osdag design)

5.6 Bolt Spacing Check

- Pitch = 67.5 mm ($2.5d$)
- Edge distance = 55 mm

5.7 End Plate Design

5.7.1 Plate Thickness

$$t_p = 56 \text{ mm} \quad (5.12)$$

5.7.2 Moment Capacity

$$M_{ep} = 11.57 \text{ kNm} \quad (5.13)$$

$$M_{dp} = 12.03 \text{ kNm} \quad (5.14)$$

Safe since $11.57 < 12.03$

5.8 Member Capacity Check

$$T_d = 1700 \text{ kN} > 223.68 \text{ kN (Safe)} \quad (5.15)$$

5.9 Comparison with Osdag

Parameter	Osdag	Manual	Difference (%)
Bolt Shear (kN)	190.8	190.8	0
Bolt Tension (kN)	306.49	306.49	0
Plate Thickness (mm)	56	56	0

5.10 Observations

- Axial load increased significantly due to IS 800 Clause 10.7.
- Prying force plays an important role in bolt design.
- End plate bending governs thickness requirement.

5.11 Bugs and Recommendations

- Load amplification is not clearly visible to the user.
- Prying force calculation lacks transparency.
- Recommendation: Provide detailed intermediate steps in output.

5.12 Conclusion

Manual verification confirms that the connection is safe and complies with IS 800:2007 provisions.

Chapter 6

Overall Observations

- Load amplification as per IS 800 Clause 10.7 significantly increases design forces.
- Manual calculations closely match Osdag outputs across all connections.
- Flange components primarily resist bending moment in moment connections.
- Weld design governs in welded connections.
- Bolt design in end plate connections is governed by tension and prying force.
- Long joint reduction affects bolt capacity in cover plate connections.

Chapter 7

Recommendations to Design Philosophy and Bugs

7.1 Lack of Transparency in Load Modification

Osdag modifies applied loads internally as per IS 800 Clause 10.7. However, this is not clearly visible in the initial stages, which may mislead users.

7.2 Hidden Intermediate Calculations

Important parameters such as:

- k_b (bearing factor)
- Prying force calculations
- Load redistribution

are not explicitly shown.

7.3 Design Philosophy Concerns

- Conservative load amplification may lead to overdesign.
- Internal assumptions are not clearly communicated.
- User has limited control over intermediate design steps.

7.4 Software Improvement Suggestions

- Display step-by-step calculations in output report
- Show load amplification clearly in input stage
- Provide explanation for reduction factors (long joint, prying, etc.)
- Improve documentation for better understanding of design logic

Chapter 8

Conclusion

Manual verification confirms that all connections designed using Osdag satisfy IS 800:2007 provisions. The study also highlights the importance of understanding underlying assumptions in software-based design and suggests improvements for better transparency.